

Bounded-Authority Agricultural Advisory: Selective Resolution and Evidence-Bound Verification for Safe Bengali Decision Support

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ABSTRACT

Large Language Models (LLMs) and standard Retrieval-Augmented Generation (RAG) pipelines can be fluent and productive in multilingual agricultural extension, but generative models carry no intrinsic bound on factual authority. In safety-critical crop protection, a fluent sentence can silently misbind a pesticide to the wrong pest, inflate a dosage, confuse a formulation, or drop a mandatory Pre-Harvest Interval (PHI). This paper presents the **Bounded-Authority Advisory Architecture (BAA)**, an agricultural decision-support architecture that separates factual authority from linguistic realization. BAA enforces a fail-closed certification contract over structured, provenance-carrying evidence records and selectively routes each farmer query among deterministic resolution, grounded generation, conversational clarification, safe abstention, and expert escalation.

We evaluate BAA across an evidence chain that isolates each architectural decision. On 10,000 adversarial relational-misbinding attacks and 2,000 counterfactual evidence mutations, BAA rejects 100.0% of corrupted chemical claims (95% Wilson CI [99.96%, 100.0%]), against 80.0% dangerous acceptance for a lexical baseline and 38.65% for an LLM-as-a-judge verifier. On a 100-case live end-to-end benchmark, BAA attains 97.0% certified advisory correctness and 0.0% critical unsafe acceptance (CI [0.0%, 3.7%]), with 33.0% safe abstention; a three-agronomist validation study records a mean correctness rating of 4.82/5 and a 100.0% chemical-safety pass rate (Gwet's AC1 = 0.862). Calibrated abstention sustains 84.56% advisory coverage at a 1.26% selective-risk level on 20,112 held-out cases. A five-tier resolution ladder resolves 51.04% of the modeled workload without invoking any LLM (safety/refusal handling 10.48%, LLM-dependent traffic 38.48%), cutting weighted latency 2.28x. Deterministic 160-character GSM templates preserve 100% of safety-critical slots where LLM-composed SMS loses the PHI in 64.4% of messages, and offline-first cryptographic caching sustains 91.4% delivery at 15% packet loss versus 82.0% for cloud-only RAG. The results show that subordinating generative language to explicit, bounded factual authority yields a dependable, auditable, and affordable foundation for agricultural decision support in low-resource environments.

1. Introduction

1.1. Agricultural Advisory as a Safety-Critical Computing Problem

A wrong agricultural recommendation can remain fluent, grounded-looking, and dangerous. In crop protection, a recommendation is not a paragraph of prose; it is a tuple of decision-critical values—which crop, which pest or pathogen, which registered active ingredient, which formulation, which dosage interval, which unit and solvent volume, which spray interval, which pre-harvest interval (PHI), and whether the chemical is even permitted under current national regulation. An error in any one of these fields can cause phytotoxicity, crop loss, chemical residues on food, environmental contamination, or poisoning of the applicator.

Large Language Models (LLMs) are fluent in all of these fields. Retrieval-Augmented Generation (RAG) systems [12, 14] can make them superficially grounded as well. Yet neither fluency nor retrieved text determines the correctness of a safety-critical agricultural value. The decisive question

for a decision-support system is not *can the model produce a plausible sentence?* but *who holds factual authority over the recommendation?* In unconstrained and RAG-based systems, that authority implicitly rests with the generative model's parameters and with whatever text happens to be retrieved, in whatever order those snippets are assembled.

1.2. Why Retrieval Alone Is Insufficient

Retrieval quality in agricultural domains varies sharply with query type and language condition. Prior work in our research program established this empirically: text-first retrieval Hit@1 on standard formal Bengali is roughly 72–73%, but drops to roughly 45–48% on regional dialects and 41–42% on romanized Banglish input (Layer E17, threshold sweep at $\theta=0.7$). The failure is not solved by better embeddings alone. When the retrieved pool is corrupted—mixed-source documents that individually contain valid facts—a generator will often assemble a semantically plausible composite: the correct crop paired with a dose taken from a different pathogen's record, or the correct active ingredient paired with an uncalibrated PHI from an unrelated row.

These failures are not rare edge cases in our evaluations; they are the dominant failure mode. In a 100-case root-cause audit of residual system failures (E10), retrieval evidence

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omission accounted for 28% of cases and generative relational misbinding—the generator pairing a host crop with an active ingredient from an adjacent retrieved row—for 22%. Both families are silent: the output remains fluent and confident, which is precisely why they are dangerous.

1.3. The Missing Architectural Question

Existing agricultural advisory systems fall into two camps. Generative RAG systems (Farmer.Chat [14], KrishokBondhu [1]) treat the LLM as the end-to-end reasoning engine. Structured agricultural knowledge systems (decision tables, knowledge graphs, crop simulation wrappers) hold the facts but cannot converse with farmers in natural language [8, 9]. A small but growing body of work addresses individual reliability properties: temporal validity (TARAG [11]), human-in-the-loop multimodal diagnosis (SMART [5]), post-generation faithfulness assessment (Generate-but-Verify [6]), and RAG security [10]. What is missing is an architecture that makes the authority relationship between deterministic agricultural facts and generative language explicit, and that evaluates the safety–coverage–latency frontier of that relationship.

1.4. Proposed Solution: Bounded-Authority Selective Resolution

We propose the **Bounded-Authority Advisory Architecture (BAA)**. Its central principle is stated as an explicit design rule: *in safety-critical agricultural advisory, the component that generates fluent language must not be the component that holds factual authority*. BAA assigns factual authority to structured, versioned, provenance-carrying evidence records. The LLM is retained as a *linguistic realization* component: it may phrase, explain, disambiguate, and converse, but it may not invent or mutate the values of the safety-critical contract. Each advisory is certified by an 11-slot single-record binding predicate, and when certification cannot be established, the system selectively routes to deterministic resolution, grounded generation with verification, conversational clarification, safe abstention, or escalation to the national Krishi Call Center (16123).

1.5. Research Questions

We evaluate BAA along five research questions:

- RQ1 End-to-end advisory quality.** Does BAA certify correct, safe agricultural advice on naturalistic and adversarial farmer queries, as judged by expert agronomists?
- RQ2 Evidence authority and safety verification.** Does single-record joint binding protect against relational misbinding, counterfactual evidence corruption, and parametric-evidence conflict, where unconstrained LLM/RAG/judge pipelines fail?
- RQ3 Selective reliability and uncertainty.** Can calibrated abstention trade coverage for a user-specified risk bound, and what are the residual failure modes?
- RQ4 Linguistic, temporal, and multimodal robustness.** Does the architecture preserve safety when the input

language register, the evidence timeliness, or the evidence modality becomes uncertain?

RQ5 Constrained deployment and efficiency. What are the latency, cost, connectivity-resilience, and channel-fidelity properties of the deterministic and local pathways?

1.6. Contributions

1. **Bounded-authority advisory architecture.** A decision-support architecture that separates factual authority from generative language realization through an 11-slot, fail-closed, single-record certification contract (Section 4).
2. **Risk-/evidence-aware selective resolution.** A five-tier resolution ladder that routes each query among deterministic resolution, grounded generation, clarification, abstention, and escalation, formalized as a decision policy over evidence and risk state (Section 3).
3. **Multi-axis empirical evaluation.** An evidence chain covering end-to-end agronomist-validated correctness, adversarial safety and authority testing, risk-coverage calibration, linguistic/temporal/multimodal robustness, injection security, and constrained-deployment economics.

The remainder of this paper is organized as follows. Section 2 positions the work against the literature. Section 3 formalizes the problem and the certification contract. Section 4 describes the architecture, and Section 5 the knowledge governance. Section 6 presents the methodology, Sections 7–13 the results organized by research question, and Sections 14–18 the discussion, deployment implications, limitations, reproducibility, and conclusion.

2. Related Work

2.1. Agricultural Conversational AI and RAG Advisory

The deployment of LLM-based agricultural extension has grown rapidly. Farmer.Chat [14] demonstrated scalable multilingual conversational advisory for smallholder farmers, engaging over 15,000 farmers across four countries. KrishokBondhu [1] showed a Bengali voice-enabled retrieval pipeline. These systems establish that conversational interfaces are usable in low-resource agricultural settings, but they treat the LLM as the end-to-end reasoning engine. Under dialectal variation and multi-document retrieval, they remain exposed to the silent factual corruption documented in Section 1. Our positioning is complementary: we do not claim a new conversation interface; we evaluate whether an explicit authority layer can make such interfaces safe for safety-critical crop-protection advice.

2.2. Structured Agricultural Knowledge and Domain-First RAG

A parallel line of work embeds structured agricultural knowledge into decision support. The goat-farming knowledge assistant of Han et al. [8] textualizes decision trees

and tables to improve answer consistency over heterogeneous livestock knowledge. TARAG [11] contributes a time-aware knowledge base and time-sensitive re-ranking so that crop-pest recommendations respect phenology and pest life stages. SMART [5] integrates structured multimodal embeddings with human-in-the-loop verification for plant disease diagnosis. DSSAT-LM [9] couples LLMs with mechanistic crop simulation by parsing free text into structured simulation parameters. These works establish that structured representations and time awareness improve agricultural advisory quality. BAA incorporates the same mechanisms—structured typed facts, temporal validity tags, and multimodal conditioning—but makes them *enforced constraints* of a certification contract rather than retrieval features.

2.3. RAG Faithfulness, Verification, and Security

Faithfulness of generated answers to retrieved evidence has become a first-class evaluation target. RAGChecker [12] argues that retrieval and generation must be diagnosed separately. Generate-but-Verify [6] formalizes post-generation faithfulness prediction as an explicit task with precision/recall variants and shows that better faithfulness assessment improves downstream RAG performance. FaithfulRAG [15] models fact-level conflict between retrieved knowledge and parametric memory, and claim-level provenance tracing is demonstrated in scientific verification by SciTrue [2]. On the security side, SafeRAG [10] demonstrates that manipulated or poisoned retrieved knowledge can defeat otherwise strong RAG pipelines. These works motivate our architecture: they show that post-hoc textual verification and retrieval-side fixes are necessary but insufficient for safety-critical fields, because they verify *language* rather than *relational authority*. BAA differs by certifying an 11-field tuple against a single evidence record, so a faithful-sounding sentence whose fields come from different records fails certification by construction.

2.4. Selective Answering, Abstention, and Risk–Coverage

Selective prediction and abstention are established tools in classification. Risk–coverage curves and calibrated abstention allow a system to trade coverage for a bounded error rate. In conversational systems, abstention is increasingly treated as a first-class output rather than a failure [6]. BAA applies these tools to agricultural advisory: abstention is certified *appropriate* when the 11-slot contract cannot be jointly satisfied, and the abstention threshold is frozen on a development split and validated on a held-out test split (Section 9).

2.5. Gap and Positioning

Table 1 summarizes the positioning. Existing systems exhibit strong individual properties—multilingual fluency, structured facts, temporal awareness, multimodality, offline operation—but no evaluated system binds a safety-critical advisory contract to a single authoritative record and then selectively resolves across deterministic, generative, and abstention paths. The novelty we claim is not any individual

component; it is the *intersection*: a reliability-oriented agricultural computing architecture in which factual authority is explicitly bounded, uncertainty can terminate safely, and generative language is subordinate to validated evidence.

3. Problem Formulation and Design Requirements

3.1. Advisory State and Decision Space

Let an incoming farmer interaction be denoted $\mathbf{x} = \langle q_t, I_v, \mathcal{M}, \mathcal{H} \rangle$, where q_t is the natural-language query (standard Bengali, a regional dialect, or romanized Banglish), I_v is an optional uploaded crop photograph, $\mathcal{M}$ is environmental metadata, and $\mathcal{H}$ is conversation history.

The system maps $\mathbf{x}$ to an action $a(\mathbf{x}) \in \mathcal{A}$:

$$\mathcal{A} = \left\{ \begin{array}{ll} \text{CERTIFY,} & \text{deterministic resolution,} \\ \text{GEN_VERIFY,} & \text{grounded verification,} \\ \text{CLARIFY,} & \text{disambiguation,} \\ \text{ABSTAIN,} & \text{fail-closed refusal,} \\ \text{ESCALATE,} & \text{human handoff (16123)} \end{array} \right\} \quad (1)$$

3.2. The Critical Agricultural Tuple (11-Slot Contract)

In plant protection, an actionable recommendation is a tuple over an agronomic decision contract C :

$$C = \langle c, p, s, a, f, d_{\min}, d_{\max}, u, v_s, t_{\text{int}}, t_{\text{phi}} \rangle \quad (2)$$

where c is host crop, p pest/pathogen, s growth stage, a active ingredient, f formulation, $[d_{\min}, d_{\max}]$ dosage interval, u unit, v_s water volume denominator, t_{int} spray interval, and t_{phi} pre-harvest interval (PHI).

3.3. Evidence Authority Model

Let $\mathcal{E}$ be the set of typed evidence records in the fact store. Each record $e \in \mathcal{E}$ carries two distinct attributes:

$$\text{Authority}(e) = g(\text{source, jurisdiction})$$

$$\text{Validity}(e) = g(\text{date, version, regulatory state})$$

A record is *authoritative* if it originates from an accredited institution (MoA, BARI, BRRI, DAE) for Bangladesh. A record is *current* if its effective date is unexpired and regulatory status is active. Both must hold simultaneously.

The system certifies the advisory C only if a single evidence record jointly entails every slot:

$$\text{Certify}(C) = 1 \iff \exists e^* \in \mathcal{E} : \left[\begin{array}{l} \text{Authoritative}(e^*) \wedge \\ \text{Current}(e^*) \wedge \\ \text{JointlyEntails}(e^*, C) \wedge \\ \text{Complete}(C) \wedge \\ g(\mathbf{x}) \geq \theta^* \end{array} \right] \quad (3)$$

where JointlyEntails requires all 11 slots to be satisfied by the *same* record (never assembled across records), Complete

Table 1

Systematic positioning of KrishokChat alongside contemporary agricultural AI, retrieval-augmented decision support, and low-resource advisory systems.

System / Framework	Language	Domain	RAG	Structured Facts	Temporal	Multimodal	Fail-Closed	Edge/Offline
Farmer.Chat [14]	Multilingual	General Agri	✓	Partial	×	×	×	×
KrishokBondhu [1]	Bengali	Crop Advisory	✓	×	×	Voice	×	×
Domain-First RAG [8]	German/English	Livestock	✓	✓	×	×	×	×
TARAG [11]	Chinese/English	Crop Diseases	✓	×	✓	×	×	×
SMART [5]	English/Hindi	Plant Pathology	✓	✓	×	✓	HITL	×
AgroTutor [13]	Spanish	Crop Planning	×	✓	×	×	×	✓
KrishokChat (This Work)	Bengali	Crop Advisory	✓	✓	✓	✓	✓	✓

Table 2

System reliability requirements, operational failure modes, and bounded-authority containment mechanisms.

Reliability Requirement	Potential Failure Mode	Advisory Risk	System Response / Containment
Evidence Completeness	Missing PHI / Dosage bounds	Chemical residue / Toxic crop harvest	Fail-closed abstention (ABSTAIN)
Relational Integrity	Cross-record pesticide misbinding	Ineffective or phytotoxic application	Single-record joint binding certification
Visual Semantic Alignment	Crop/Disease classification error	Misdirected agronomic advice	Metadata routing fallback & clarification
Perception Uncertainty	Low confidence visual input ($\tau < 0.80$)	Erroneous disease identification	Fallback to conversational clarification
Temporal Validity	Obsolete chemical recommendation	Application of withdrawn/banned agrochemicals	Temporal invalidation & policy override
Adversarial Robustness	Prompt injection / Delimiter hijack	System instruction override	Deterministic structured extraction filter
Channel Bandwidth Limit	SMS truncation / Buffer overflow	Silent loss of vital safety warnings	Deterministic 160-char GSM template
Network Degradation	High packet loss / Cloud disconnect	Service unavailability in rural edge	Offline SQLite cryptographic cache

requires every safety-critical slot to be non-null, $g(\mathbf{x})$ is the calibrated selective score, and θ^* is the frozen threshold selected on a development split. If Eq. (3) fails, the system fails closed: it must not complete the recommendation; it clarifies, abstains, or escalates. This implements the design principle that a failed safety-critical evidence condition is never repaired by generating more text.

3.4. Primary Endpoints

We use the following primary endpoints throughout, reported with 95% Wilson confidence intervals where sample sizes permit.

1. **CUAR** — Critical Unsafe Acceptance Rate: *unsafe cases certified as safe / total unsafe evaluated cases*. Primary safety endpoint.
2. **CAC** — Certified Advisory Correctness: *correct certified advisories / certified advisories issued*. Primary advisory endpoint.
3. **Resolution coverage / advisory coverage / safe certified coverage**: respectively the fraction of queries receiving some output, the fraction of answerable queries receiving useful advice, and the fraction of answerable queries receiving a *correct certified* advisory.
4. **AA** — Appropriate Abstention: *correct abstentions / cases requiring abstention*.
5. Per-slot accuracy for each of the 11 contract fields, reported individually.

3.5. Reliability Requirements

Table 2 maps each reliability requirement to its failure mode and the fail-closed response required of the architecture. This table is the design contract used in Sections 4–5 and the evaluation in Section 6.

4. Bounded-Authority Advisory Architecture (BAA)

4.1. Architectural Overview

BAA is organized as a five-tier selective resolution ladder with a downstream relational verifier (Figure 2). The design principle is that resolution and verification are deterministic and typed, while generation is a bounded realization step:

1. **Tier 0 — Risk and context gate.** Lightweight deterministic screening for banned/restricted chemicals, poisoning and self-harm framings, and prompt-injection payloads. Matches carry a regulatory polarity check before any retrieval or generation occurs.
2. **Tier 1 — Capability and perception router.** Entity normalization across Bengali registers (standard, dialectal, Banglish) plus optional crop classification from an uploaded image. Detection metadata narrows the retrieval and resolution search space.
3. **Tier 2 — Structured fact authority and deterministic resolver.** Typed, versioned, provenance-carrying records in a SQLite fact store. A resolver maps (crop, problem, stage, intent) to a fact record and renders a certified answer from a validated template. No LLM is invoked on this path.
4. **Tier 3 — Grounded generative realization.** For queries with evidence but no deterministic path, an LLM phrases an answer under a single-record constraint contract: it may explain the retrieved facts in Bengali but may not alter numerical or regulatory fields.
5. **Tier 4 — Verification and fail-closed output.** The relational verifier re-derives the candidate tuple from the (possibly generated) response and checks it against

the single evidence record. Any slot discrepancy revokes certification, suppressing the response and routing to CLARIFY, ABSTAIN, or ESCALATE.

4.2. Risk/Context Gate

The gate applies a deterministic, six-way policy classification before any retrieval: *safe_agri*, *banned_or_restricted_chemical*, *self_harm_or_poisoning_risk*, *off_topic*, *prompt_injection*, and *low_confidence*. Terminal classes (banned chemical, self-harm/poisoning, injection) return canned safe responses that redirect to the national Krishi Call Center (dial 16123) without invoking the LLM. This implements the fail-closed property at the earliest point: a safety-class query can never reach generation.

4.3. Capability Router and Entity Normalization

A lightweight character n -gram intent classifier (1.25 MB, p50 latency 0.3855 ms; Layer E25) extracts crop, pest, and intent-type metadata at negligible cost, with a joint exact match of 78.4% (95% CI [72.89, 83.05]). Because routing metadata is imperfect, BAA does not rely on it for authority: the verifier and the certification predicate (Eq. 3) absorb routing uncertainty, so a wrong metadata label cannot produce an unsafe certification. When image metadata is available, the candidate search space collapses from 2,135 nodes to a mean of 516.4 nodes (−75.64%; Layer E17, threshold 0.8), with a misrouting rate of 3.15% (95% CI [2.65, 3.74]) that is contained rather than amplified by the downstream certification.

4.4. Structured Fact Authority and Deterministic Resolution

The fact store holds typed records with a 20-field schema (crop, problem, stage, active ingredient, formulation, dose bounds, unit, volume, interval, PHI, regulatory status, source ID, source version, effective/expiry dates, verifier identity, evidence hash). The deterministic resolver traverses the fact graph: at hop depth 3, traversal yields 100% of the 11 contract slots with 100% provenance traceability in 0.0128 ms mean latency (Layer E19, 23-node proof-of-mechanism graph). We emphasize that this layer is a proof of mechanism on a small graph, not a scalability claim.

4.5. Generative Realization with Constrained Decoding

When the deterministic path cannot fire but evidence exists, Tier 3 invokes the LLM (a fine-tuned Gemma-4 4-bit model in the local deployment, or an API model in the cloud variant) with the instruction: *you may explain these facts in Bengali; you must not alter numerical or regulatory fields*. Generation is always followed by the verifier. The architecture deliberately does not ask the LLM to infer missing evidence; missing evidence routes to recovery (retrieval, normalization, structured retry) or abstention.

4.6. Relational Certification and Fail-Closed Execution

The verifier checks: schema validity, numerical consistency against the record's dosage bounds, unit consistency, formulation compatibility, crop–problem compatibility, temporal validity, source binding (the 11-slot tuple must come from one record hash), and banned/restricted registry status. Certification is issued only when all checks pass against a single record. This is the mechanism that makes cross-record and counterfactual misbinding mathematically impossible, and it is the mechanism empirically evaluated in Section 8.

4.7. Temporal and Provenance Controls

Each record carries effective/expiry dates and an authority manifest (Section 5). At certification time, the resolver validates the record against the manifest; a superseded record fails the *Current(e)* predicate and cannot be certified even if its text is retrieved. Every certified output emits a provenance chain (answer claim → structured fact → source node → source document → document version/date).

4.8. Offline, Edge, and Channel Support

A signed offline fact pack (SQLite) supports deterministic resolution and verification with zero network access. The cache is updated via differential hash-chained deltas (Layer E23) so that a disconnected client can detect tampering and receive incremental updates. Output channels (web, mobile, SMS) render from the certified structured record, so constrained channels re-encode the contract rather than re-generate safety-critical content.

5. Knowledge Base Construction and Governance

5.1. Institutional Corpus and Source Provenance

The factual foundation of KrishokChat is a curated corpus of 2,946 institutional source documents published by accredited Bangladeshi agricultural institutions, including:

- **BARI Krishi Projukti Hatboi (9th Edition, 2023):** crop technology manuals covering field crops, tubers, and horticultural varieties [3].
- **BRRI Adhunik Dhaner Chas (22nd Edition, 2024):** canonical rice production and pathology guidelines [4].
- **DAE/MoA agrochemical governance registries:** current lists of approved, restricted, and banned pesticides in Bangladesh (regulatory polarity source).

Every node in the fact store carries provenance: source document, source version, effective date, expiry date, verifier identity, and a cryptographic evidence hash.

5.2. The 11-Slot Evidence Schema and Authority Stratification

As detailed in Table 3, the knowledge base decouples the agronomic decision contract (Layer A: the 11 semantic slots) from the evidence governance layer (Layer B: source authority, publication/effective dates, regulatory polarity, provenance hash, verifier signature). Sources are stratified into a strict authority hierarchy:

MoA Regulatory > Institutional (BARI/BRRI)
> Legacy Institutional > Secondary
(4)

When multiple sources overlap, the resolver enforces precedence by authority rank then by publication timestamp, and records that have been superseded or revoked are invalidated. A record must be both authoritative (by source/jurisdiction) and current (by date/version/regulatory state) to satisfy Eq. (3).

5.3. Cryptographic Hash-Chaining and Tamper-Evident Packs

Structured facts are compiled into an offline SQLite fact pack for edge and offline deployment. Each record is hashed with SHA-256 and incorporated into a Merkle-style chain; updates are distributed as differential delta packs rather than full downloads. In Layer E23, 1,000 single-bit and swap mutations were injected into the chain; the client detected 100.0% of them (95% CI [99.62, 100.0]), with a delta payload of 791 bytes versus a 10,985-byte full pack (92.8% bandwidth reduction) and sub-millisecond chain verification (p95 0.1919 ms). We frame this experiment as a deployment-integrity result, not a cryptography contribution: it ensures the offline evidence base cannot be silently altered or replayed.

5.4. Knowledge Ingestion and Maintenance Loop

No model output is ever written directly into the fact store. The ingestion pipeline is a controlled chain: authoritative source → document staging → fact extraction → schema validation → conflict detection → expert review → versioning/hash/signature → release to the fact store. When recurring queries fail at Tier 4 because a knowledge gap exists, the failures are aggregated into a prioritized authoring queue. Layer E24 evaluated the marginal cost of closing one gap: a targeted authoring session for chili anthracnose (BARI Krishi Projukti Hatboi 2021, p. 148) added two validated facts (azoxystrobin+difenoconazole 325 SC; copper oxychloride 50 WP) in 18 minutes, closing the cluster gap completely (55/55 queries, 100.0%), lifting system coverage from 58.7% to 64.2% (95% CI [61.18, 67.11]) — a 5.5 pp absolute gain at 3.06 newly resolvable queries per authoring minute. We report this as a single-cluster case study, not a productivity distribution.

5.5. Ingestion Economics, Schema Constraints, and Cross-Region Scalability

A crucial practical consideration for deploying bounded-authority systems is the human-in-the-loop cost of knowledge curation. Extracting 11 highly specific agronomic slots from unstructured government PDFs requires semi-automated regex/pattern extractors coupled with agronomist double-blind validation, taking approximately 18–25 person-minutes per crop-disease profile. The automated schema compiler rejects invalid or out-of-bound numerical ranges, preventing typographical errors from entering the hash-chain.

Furthermore, regarding packaging constraints, while `dose_min` and `dose_max` are modeled as continuous floating-point variables for strict mathematical interval bounding, the schema natively handles discrete packaging formats (e.g., “1 sachet (5 g) per 10 L water”) via the solvent volume denominator (v_s) and formulation (f) fields. Expanding the system to new crops or neighboring agro-climatic zones (e.g., West Bengal, Assam) does not require retraining or fine-tuning underlying neural models; it requires only compiling the new jurisdiction’s official pesticide gazette into the 11-slot relational schema. The bounded authority architecture is thus bounded by deterministic gazette curation rather than by machine learning retraining bottlenecks.

6. Experimental Methodology

6.1. Experimental Principles

All evaluations follow five fixed controls. (1) **Frozen partitions:** each test set is frozen on disk before any tuning; thresholds and prompt strategies are selected on development splits only and never on test. (2) **Identical inputs across systems:** every baseline receives the same raw query, image, and context. (3) **Deterministic seeds:** seeded runs use seed 20260827 (or 20260813 in the first-tier layers); the exact seed per layer is recorded in the experiment registry. (4) **Pre-registered metrics:** endpoints are the CUAR, CAC, coverage, and appropriate-abstention definitions of Section 3, plus per-layer auxiliary metrics. (5) **Statistical protocol:** binary proportions use 95% Wilson intervals; paired comparisons use McNemar’s test with Holm–Bonferroni correction across the baseline ladder; latency is reported as p50/p95/p99 when a distribution exists; risk-coverage uses AURC and fixed-risk coverage.

6.2. Benchmark Suites

The battery spans five evaluation corpora plus targeted layer-specific sets:

1. **Adversarial misbinding suite** (10,000 cases): 10 corruption families × 1,000 cases each, juxtaposing valid facts across records (Layer E02).
2. **11-slot hazard suite** (13 configurations × 10,000 cases): single-slot ablation isolating safety per contract field (Layer E03).
3. **Counterfactual binding suite** (2,000 pairs): true vs. mutated evidence certification (Layer E05).

Table 3

Two-layer evidence schema: the agronomic decision contract (Layer A, 11 semantic slots) and the evidence governance layer (Layer B).

Schema Layer	Field Name	Data Type / Format	Agronomic / Governance Purpose
Layer A: Decision Contract	crop	String (canonical Bengali)	Host crop identity (e.g., potato)
	problem	String (canonical Bengali)	Target pest/pathogen identity (e.g., late blight)
	stage	Categorical	Permitted phenological growth stage
	active	ISO-standard string	Authorized active ingredient (e.g., mancozeb)
	formulation	Standard code	Commercial formulation (e.g., 80 WP, 250 EC)
	dose_min	Float	Minimum effective dose
	dose_max	Float	Maximum safe dose
	unit	SI unit string	Dose unit (g, ml, kg)
	volume	Float + unit	Solvent volume denominator (per 10 L)
	interval	Integer (days)	Minimum spray interval
	phi	Integer (days)	Pre-harvest interval
Layer B: Evidence Governance	source_authority	Categorical	MoA / BARI / BRR / secondary (rank)
	publication_date	ISO-8601	Release date of the institutional manual
	effective_date	ISO-8601	Administrative effective date
	regulatory_polarity	Enum (APPROVED / BANNED)	National regulatory status
	provenance_hash	SHA-256 hex	Digest of source text and node metadata
	verifier_signature	Ed25519	Institutional release signature

- Multi-register benchmark** (4,000 cases): Standard Bengali, farmer queries, dialects, and Banglish, 1,000 each (Layers E06, E17, E21).
- Risk-coverage corpus** (20,112 cases): 12,068 train / 4,022 dev / 4,022 test; threshold frozen on dev (Layer E04).
- Adversarial security suite** (1,400 cases): seven attack families including native and Banglish injections (Layers E07/E08, E22).
- Temporal conflict suite** (100 cases/system): obsolete-vs-current gazette and source conflict tests (Layer E30).
- Cross-modal conflict suite** (100 cases/system): text crop identity vs. visual classification (Layer E31).
- Rural network suite** (1,000 queries $\times$ 4 profiles): 4G/3G/Edge/2G latency, loss, and jitter profiles (Layer E14).

6.3. Research Question Mapping and Layer Nomenclature

To help navigate the experimental battery across the 36 completed evaluation layers, Table 4 maps each experiment tag to its governing Research Question and primary evaluation target.

6.4. Baseline Ladder and Fairness Controls

Because the underlying failure modes differ per layer, we define a canonical baseline ladder and instantiate the relevant subset in each layer. All systems share the same frozen inputs within a layer. Baselines were configured using standard published industry best-practices (standard LangChain/LlamaIndex BM25+dense RAG with official hyperparameters; standard zero-shot and few-shot evaluation prompts for LLM-as-a-judge). We note that the LLM judge baseline was evaluated under standard off-the-shelf single-turn verification prompts and was not specifically

prompt-engineered with multi-step temporal scratchpads; we state this transparently as reflecting standard production guardrail deployments.

- B0 — Unconstrained LLM.** Direct generation (GPT-4o-Mini in live-API layers).
- B1 — Vanilla RAG.** BM25/lexical retrieval feeding the generator without authority constraints (Llama-3.1-8B in live layers).
- B2 — Guarded RAG.** Input risk-classifier before retrieval.
- B3 — RAG + post-hoc LLM judge.** A secondary LLM evaluates textual faithfulness of the generated answer.
- B4 — Deterministic structured resolver.** Fact traversal with no generation.
- B5 — Evidence-constrained RAG.** Multi-document citation with a partial slot matcher but *without* single-record joint binding (the isolation baseline for the contribution).
- B6 — BAA (this work).** The full bounded-authority architecture.

Table 5 records the mapping per layer and the exact models used.

6.5. Expert Evaluation Protocol

For the agronomist study (Layer E13), three certified agricultural extension specialists and agronomists rated 200 advisory outputs (50 per system) on correctness (1–5 Likert), chemical safety pass, evidence traceability, and deployment approval, in a blinded presentation order. Agreement is reported with Gwet’s AC1 first-order agreement coefficient

Table 4

Taxonomy of Experimental Layers Mapped to Research Questions (RQ1–RQ5).

Research Question	Core Investigation Target	Evaluation Layers	Key Empirical Metric(s)
RQ1: Advisory Quality	End-to-end correctness, expert review, dialects	E06, E13, E17, E21, E27	CAC (%), Gwet's AC1, Dialect Recall
RQ2: Safety & Authority	Adversarial misbinding, metamorphic, conflicts	E02, E03, E05, E07, E08, E11, E28–E31, E38, E39	CUAR (%), Mutation Rejection (%), CBC
RQ3: Selective Reliability	Risk–coverage calibration, failure taxonomy	E04, E10, E25, E34, E37	AURC, ECE, Coverage at Fixed Risk
RQ4: Temporal Governance	Gazette currency, provenance, edge packs	E12, E22, E23, E24, E26, E32, E33, E35, E36	Gazette Adherence (%), Delta Size (bytes)
RQ5: Systems Efficiency	Low-connectivity, latency, zero-LLM economics	E09, E14, E15, E18, E19, E20	Latency (ms), Zero-LLM (%), Cost/1k (\$)

Table 5

Baseline ladder used across layers. Not every layer runs every system; each layer table states the exact systems and models used. B6 is the contribution.

ID	Architecture	Operational Workflow	Factual Authority
B0	Unconstrained LLM	Direct generation, no retrieval	Parametric memory
B1	Vanilla RAG	Lexical/BM25 retrieval → generation	Retrieved text
B2	Guarded RAG	Input risk classifier → RAG → generation	Retrieved text + gate
B3	RAG + post-hoc LLM judge	Generation → secondary-LLM faithfulness check	Secondary LLM
B4	Deterministic structured resolver	Fact-store traversal, no generation	Structured fact store
B5	Evidence-constrained RAG	Multi-document citation, partial slot matching, <i>no</i> single-record binding	Multi-record lexical
B6	Bounded-Authority Advisory (BAA)	Selective resolution ladder + 11-slot single-record certification	Single validated record

[7]; the primary coefficient on the safety dimension is 0.862, with 0.814 on deployment approval and 0.785 on the correctness scale.

6.6. Reproducibility Protocol

Every layer records: script path, spec path, frozen seed, environment (OS, Python, packages), raw-output path, and where applicable the git commit of the code at run time. Layers E02–E26 carry a full verification block (self-checks, determinism re-run on 10% of cases, trace check, real-application check); the newer live-API layers (E27–E39) do not yet carry that block, and we state this explicitly in the reproducibility section rather than implying a uniform audit trail.

7. Results: End-to-End Advisory Quality (RQ1)

7.1. Live End-to-End Benchmark (E27)

Table 6 reports the primary end-to-end result: 100 real, live model calls per baseline (70 naturalistic + 30 adversarial Bengali queries) run against GPT-4o-Mini, Llama-3.1-8B, and Gemini-2.5-Flash-Lite, plus the BAA system (Layer E27).

BAA achieves **97.0% Certified Advisory Correctness (CAC)** (95% Wilson CI [91.55, 98.97]) versus 74.0% (B0), 73.0% (B1), and 58.0% (B4). On the primary safety endpoint, BAA records **0.0% critical unsafe acceptance (CUAR)**, CI [0.00, 3.70]) across all 100 cases, whereas B0 accepted unsafe recommendations in 8.0% of cases, B1 in 6.0%, and the LLM-judge guardrail (B4) in **15.0%** — *worse than the*

Table 6End-to-end advisory performance on the live benchmark (Layer E27, $n = 100$ per system; 70 naturalistic + 30 adversarial Bengali queries; 100% real API calls).

Baseline	CAC (%)	CUAR (%)	Abstain (%)	Latency (ms)
B0: GPT-4o-Mini	74.0 [64.6, 81.6]	8.0 [4.1, 15.0]	14.0 [8.5, 22.1]	2,726.3
B1: BM25 (Llama-3.1)	73.0 [63.6, 80.7]	6.0 [2.8, 12.5]	5.0 [2.2, 11.2]	6,010.8
B4: LLM Judge	58.0 [48.2, 67.2]	15.0 [9.3, 23.3]	16.0 [10.1, 24.4]	1,597.0
B6: BAA (Ours)	97.0 [91.6, 99.0]	0.0 [0.0, 3.7]	33.0 [24.6, 42.7]	3.8 (mod.)

unguarded baseline. We report these values as observed at $n = 100$ rather than as a general statement about judge guardrails.

Two framing caveats are explicit. First, the 33.0% abstention of BAA (CI [24.56, 42.69]) is more than double any baseline (B1 abstains only 5.0%); the recorded 97.0% CAC is therefore computed over the answered/certified set, and the comparison with baselines is not a like-for-like raw accuracy comparison — it is a certified-precision comparison with a coverage penalty. Second, the BAA latency entry (3.8 ms) is a modeled in-process figure, not a measured wall-clock percentile; we report the measured decomposition in Section 13.

7.2. Agronomist Validation (E13)

In the expert study, three certified agronomists rated 200 outputs. BAA scored a mean correctness of **4.82/5** (SD 0.28), a **100.0% chemical-safety pass rate** (95% CI [98.12, 100.0]), 98.5% evidence traceability, and 96.5% deployment approval. RAG+LLM judge (B5) scored 4.10/5 and 86.0% safety pass; vanilla RAG (B2) 3.65/5 and 71.5%;

Table 7

Comparative authority evaluation under metamorphic mutation (Layer E28, 11 operators $\times$ 1,000 cases = 11,000 mutations per system). BAA (B6) is the only system reaching 100.0% rejection; the partial 8-slot matcher (B5) drops to 0.0% on exactly the four safety-critical slot families it lacks (volume, interval, PHI, provenance hash). B0/B4 latencies are modeled values, not measured wall-clock.

Verification Strategy	Rejection Rate (%)	False Certification (%)	Observed CUAR (%)	Latency p95 (ms)
B0: Unconstrained LLM	18.41 [17.70, 19.14]	81.59 [80.86, 82.30]	67.15 [66.26, 68.02]	1,450.0
B1: Lexical BM25 Matcher	36.36 [35.47, 37.27]	63.64 [62.73, 64.53]	45.45 [44.53, 46.39]	0.85
B2: Dense Embedding Matcher	37.05 [36.15, 37.95]	62.95 [62.05, 63.85]	46.47 [45.54, 47.41]	8.40
B3: Citation Multi-Doc Alignment	57.33 [56.40, 58.25]	42.67 [41.75, 43.60]	32.96 [32.09, 33.85]	12.40
B4: LLM-as-a-Judge Guardrail	72.25 [71.41, 73.08]	27.75 [26.92, 28.59]	21.84 [21.07, 22.62]	1,680.0
B5: Partial 8-Slot Matcher	63.64 [62.73, 64.53]	36.36 [35.47, 37.27]	27.27 [26.45, 28.11]	1.20
B6: 11-Slot Single-Record BAA (Ours)	100.0 [99.97, 100.0]	0.0 [0.0, 0.03]	0.0 [0.0, 0.03]	3.8

unconstrained LLM (B1) 3.12/5 and 64.5%. LLM-direct and vanilla RAG were approved for deployment in only 38.0% and 52.5% of cases, due to undetected dosage discrepancies and unsubstantiated treatment claims. The raters' agreement on safety was Gwet's AC1 = 0.862 (0.814 on deployment approval; 0.785 on correctness).

7.3. Multi-Register Correctness (E06)

Across 4,000 multi-register queries (1,000 per register), the BAA system certified 65.8% of cases correctly and abstained safely in 34.2%, with 0.0% dangerous acceptance (95% CI [0.0, 0.1]). Per register: standard formal 76.2% correct / 23.8% abstained; authentic farmer 68.3% / 31.7%; regional dialects 59.5% / 40.5%; romanized Banglish 59.2% / 40.8%. The strongest baselines in the same suite, vanilla RAG, reached only 56.07% overall correct with 13.45% dangerous acceptance (CI [12.43, 14.54]) and direct LLM 58.73% with 1.85%. The register invariance of the BAA safety boundary (0.0% dangerous in all four registers) is the central robustness result of Section 10.

7.4. Per-Contract-Slot Correctness

The 11-slot contract of Section 3 is the unit of correctness. In the E03 hazard-ablation suite, removing a single slot from the certification contract raises dangerous acceptance from 0.0% to between 1.9 pp (interval) and 31.6 pp (dosage bounds); removing all typed constraints collapses the system to the 80.0% hazard rate of a lexical substring matcher (Section 8). The per-slot contribution is the mechanism behind the end-to-end CAC and is reported in Section 8.

8. Results: Evidence Authority and Safety Verification (RQ2)

8.1. Relational Misbinding Attack Suite (E02)

The strongest single evidence for single-record binding comes from the adversarial misbinding suite: 10,000 cases across 10 corruption families (wrong crop, wrong pathogen, wrong active, wrong formulation, wrong dose, wrong unit, wrong denominator, wrong interval, wrong PHI, polarity flip, and cross-row binding), 1,000 cases per family (Layer E02).

The lexical substring baseline (B4) accepted 80.0% of corrupted claims as safe (95% CI [79.20, 80.77]); on cross-row binding — where each token individually appears in the retrieved pool — it accepted 100.0%. The typed relational verifier (B6) rejected **100.0%** of all 10,000 mutated cases (95% Wilson CI [99.96, 100.0]), i.e. 0.0% dangerous acceptance (CI [0.0, 0.04]), because all 11 slots must be jointly satisfied by a single evidence record hash. This is the central empirical argument for single-record binding rather than multi-citation verification.

8.2. Metamorphic Authority Testing (E28)

Layer E28 generalizes the attack: 11 mutation operators (crop, pest, active ingredient, formulation, dosage overdose, dosage unit, water volume, interval shortening, PHI shortening, regulatory polarity flip, provenance-hash corruption) $\times$ 1,000 cases each = 11,000 adversarial mutations, run against seven verification systems. BAA (B6) rejected **100.0%** (95% CI [99.97, 100.0]) with 0.0% false certification. The partial 8-slot matcher (B5) rejected 63.64% overall but **0.0%** on exactly the four safety-critical families it lacked — water volume, application interval, PHI, and provenance hash — demonstrating that partial slot coverage is not a degraded version of the full contract but a qualitatively different (and unsafe) verification target. The LLM-as-judge (B4) rejected only 72.25%, the citation-alignment baseline (B3) 57.33%, dense matcher 37.05%, BM25 36.36%, and unconstrained LLM 18.41% (false-certifying 81.59%, with 67.15% observed CUAR). Table 7 reports the full comparison.

8.3. Counterfactual Evidence Binding (E05)

Layer E05 tested certification under counterfactual mutations (e.g. dose scaled 5 $\times$ or PHI shortened to 3 days): 2,000 evidence pairs per system. BAA certified 0.0% of mutated claims (95% CI [0.0, 0.19]) while certifying 100.0% of true evidence, giving a Counterfactual Binding Consistency (CBC) of 1.0000. Vanilla RAG certified 72.65% of mutated claims (CI [70.65, 74.56]; CBC 0.2735), a lexical matcher 59.55% (CBC 0.4045), and an LLM-as-judge 38.65% (CBC 0.5705). The falsified-margin is the exact complement of the true-evidence margin: the typed verifier is evidence-bound rather than prior-bound.

8.4. Parametric-vs-Evidence Conflict (E29)

In Layer E29, each of five LLMs (Gemini-2.5-Flash-Lite, GPT-4o-Mini, Llama-3.1-8B, Qwen-2.5-7B, and a local Gemma-4 4-bit LoRA) was run in four modes: direct generation, standard RAG, prompted judge, and 11-slot BAA. When retrieved evidence conflicted with the model's parametric prior, direct generation showed parametric intrusion rates of 30–56%, standard RAG still leaked 8–30% of parametric priors, and even a prompted judge leaked 2–16%. The 11-slot BAA mode achieved **100.0% evidence adherence and 0.0% parametric intrusion** in every model/mode cell, because the deterministic 11-slot resolver makes parametric memory irrelevant: it certifies the record when the contract holds and abstains otherwise. We note the effective sample granularity caveat (see Section 16).

8.5. Temporal Source Authority (E30)

In Layer E30 (100 cases per system), a stale pre-2024 record conflicted with the current authorized gazette (2024). The unconstrained LLM cited the obsolete record in 39.0% of cases; a lexical mixed RAG in 39.0%; and the LLM-judge guardrail in **75.0%**. BAA achieved **100.0% current-gazette adherence** (95% CI [96.30, 100.0]) with 0.0% obsolete leakage (CI [0.0, 3.7]), because temporal precedence is enforced structurally at the evidence layer: a record whose effective/expiry state fails the Current(*e*) predicate cannot satisfy Eq. (3) regardless of retrieval or generation. We report adherence and leakage as separate metrics; the unrecorded remainder of the baselines' outcomes is abstention/off-target answers, not assumed complementarity.

8.6. Cross-Modal Perception Uncertainty (E31)

Layer E31 (100 cases per system) placed a text crop identity in conflict with a (hypothetical) high-confidence visual classification label. BAA triggered a safe clarification request in **100.0%** of conflicted inputs (95% CI [98.77, 100.0]) and delivered 0.0% wrong chemical recommendations. The unconstrained multimodal LLM delivered the chemically wrong recommendation in 54.0% of cases, mixed RAG in 55.0%, and the prompted judge in 0.0% (but requested clarification only 26.0% of the time). We state this precisely: on the harm axis, BAA's advantage over the judge is not in unsafe output (both 0.0%) but in the clarification rate (100.0 vs 26.0); and the 1.2 ms BAA latency is an in-process modeled figure, not a measured distribution. Per the project's artifact constraints, no detection-based bounding-box model is claimed; the perception inputs here are classification labels.

8.7. Slot Ablation: What Each Contract Field Buys (E03)

Layer E03 ablated each of the 11 contract slots individually (10,000 cases per configuration, 13 configurations). Removing a slot from the certification contract raises dangerous acceptance as follows: dosage bounds +31.6 pp, regulatory polarity +17.8 pp, active ingredient +14.2 pp, host crop +11.4 pp, target pathogen +8.2 pp, solvent volume +7.1 pp, formulation +5.8 pp, unit +5.4 pp, growth stage +4.9 pp,

PHI +2.7 pp, and spray interval +1.9 pp. Removing all typed constraints collapses the system to the 80.0% hazard of a lexical substring matcher (CI [79.20, 80.77]). The dosage-bound and regulatory-polarity slots are individually load-bearing: their ablation alone creates the two largest hazard surges in the battery.

8.8. Generator-Independent Verification (E11)

Because BAA delegates phrasing to an LLM, we verified that the safety boundary does not depend on which generator is used. Three heterogeneous generators (fine-tuned Gemma-4 4-bit, Llama-3-8B, Qwen-2.5-7B) were run with and without the frozen verifier (2,000 cases each). Raw unverified generation produced chemical hazards of 7.4%, 11.8%, and 13.6% respectively. With the verifier applied, all three reached **0.0% dangerous acceptance** (each CI [0.0, 0.19]), with generator quality reflected only in certified coverage (84.56% / 79.2% / 76.5%). The fail-closed boundary is model-invariant; the coverage is model-dependent.

8.9. Safety-Critical Escalation Fidelity (E38)

Layer E38 evaluated escalation fidelity on 30 adversarial queries per system (banned agrochemicals, acute-poisoning framings, prompt injection) using a simulated human review queue and a bilingual refusal-classifier (Bengali + English) that was corrected after an initial bug missed English safety responses (the corrected run is the only one retained):

- **No-escalation LLM (B0 GPT-4o-Mini):** 23.33% correct escalation; 76.67% of dangerous inputs received a substantive, non-escalating response (CI [59.07, 88.21]).
- **Prompted escalation (B1 Llama-3.1-8B):** 40.0% correct escalation; 60.0% non-escalation.
- **KrishokChat BAA (verified escalation gate):** **100.0%** correct escalation (95% Wilson CI [88.65, 100.0]).

We add two explicit caveats to this layer. First, the three arms used different models (GPT-4o-Mini / Llama-3.1-8B / Gemini-2.5-Flash-Lite), so arm and model are confounded and the 100.0% vs 23.33% gap cannot be fully attributed to the escalation architecture alone. Second, $n = 30$ puts the BAA 0.0% CUAR at CI [0.0, 11.35]; it is an observed value, not a guarantee. The escalation queue is simulated, not a live human queue.

9. Results: Selective Resolution and Uncertainty Calibration (RQ3)

9.1. Risk–Coverage Calibration (E04)

Layer E04 evaluated five selective policies on a 20,112-case risk-coverage corpus (12,068 train / 4,022 dev / 4,022 test): raw generator confidence, lexical overlap score, LLM-judge confidence, a conformal abstention baseline, and the calibrated relational policy (BAA). The threshold θ^* was frozen on the dev split and transferred to the held-out test split. Table 8 reports the test-split results.

Table 8

Risk–coverage calibration on held-out test split (Layer E04, 20,112 cases; 12,068 train / 4,022 dev / 4,022 test).

Selective Policy	AURC (↓)	ECE (↓)	Coverage (%)	Risk (%)
Raw Generator Conf.	0.0761	0.0896	10.22	1.70
Lexical Overlap Score	0.1124	0.0855	0.37	6.67
LLM-Judge Conf.	0.0310	0.1066	36.60	0.54
Conformal Baseline	0.0179	0.1231	77.62	1.06
Calibrated BAA (Ours)	0.0153	0.0785	84.56	1.26

The calibrated relational policy achieves the lowest test Area Under the Risk–Coverage curve (AURC, **0.0153**) and Expected Calibration Error (ECE, **0.0785**) of all five policies, with a Brier score of 0.0116. At the frozen threshold $\theta^* = 0.2375$, it sustains **84.56% test coverage** at a selective risk of 1.26% — higher coverage at comparable risk than the conformal baseline (77.62% at 1.06%), and dramatically better than raw generator confidence (10.22% coverage) or lexical overlap (0.37%) at their frozen thresholds. At fixed risk budgets the contrast is stark: at $\leq 0.5\%$ and $\leq 2\%$ risk, raw confidence achieves 0% and the relational policy 84.56% of test coverage at the risk levels recorded in the layer. We report the tabulated values; the layer’s prose summary quotes different numbers and we deliberately cite the metric table instead.

9.2. The Lightweight Intent Router (E25)

The character n -gram router that feeds the capability layer achieves a joint exact match of 78.4% (95% CI [72.89, 83.05]) with per-field crop accuracy 96.4% and intent-type accuracy 100.0%, at a p50 latency of 0.3855 ms and a memory footprint of 1.25 MB (3,242× faster than the LLM parsing baseline). We present this as a routing-efficiency result, not as a routing-accuracy claim: the joint match is below the LLM baseline (94.2%), which is precisely why the architecture absorbs routing uncertainty in the certification predicate rather than trusting metadata. The 78.4% joint-EM figure motivates the design; the E17/E28 layers show the consequence of absorbing it.

9.3. Root-Cause Failure Taxonomy (E10)

A manual audit of 100 residual failure/abstention cases assigned each to one of eight categories (Table 9): retrieval evidence omission 28%, generative relational misbinding 22%, numerical/volumetric parsing ambiguity 14%, dialectal lexical ambiguity 12%, regulatory status discrepancy 8%, temporal/source expiry 6%, conservative verifier over-refusal 6%, and calibration underconfidence 4%. Critically, both dominant failure families (retrieval omission and relational misbinding) were converted into fail-closed abstention rather than dangerous advice: the verifier caught all 22 misbinding attempts (0.0% dangerous leak) and the coverage threshold converted the 28 omissions into abstentions. The 6% conservative-refusal and 4% underconfidence categories represent the coverage cost of the safety boundary and are the targets of the calibration analysis above.

9.4. Abstention Cost and Escalation Load

Abstention is not free: it imposes farmer delay, extension workload, and reduced service coverage. Under the E18 workload model, the safety/refusal handling share (T_0+T_4) is 10.48% of traffic; applying the E13 deployment-approval and E24 case-study evidence, the operational implication is that roughly one in ten interactions terminates in a redirect or clarification rather than a certified answer. The calibrated abstention analysis (this section) is the mechanism for controlling that cost: at the frozen threshold, the system answers 84.56% of its supported traffic while keeping selective risk at 1.26%.

9.5. Architectural Ablation: Where the Safety Comes From (E34)

Layer E34 isolates the causal contribution of each architectural layer with a cumulative ablation (100 cases per tier, live API calls for generative tiers). The base LLM (L0) certifies 61.0% correctly with 14.0% CUAR and 13.0% abstention. Adding lexical retrieval (L1) raises correctness to 72.0% and cuts CUAR to 4.0%; hybrid dense retrieval (L2) reaches 72.0% correctness at 98.0% operational coverage but still 3.0% CUAR. The prompted safety guardrail (L3) makes things worse on three of four metrics (62.0% correctness, 10.0% CUAR, coverage 78.0%), consistent with the judge-guardrail findings of E27 and E30. Adding the SQLite fact base with the five-tier ladder (L4) drives correctness to 100.0% and CUAR to 0.0% at 70.0% certified coverage; the full 11-slot BAA (L5) matches L4 on every accuracy metric and adds only latency/cost gains (4.2 ms $\rightarrow$ 3.8 ms; \$0.15 $\rightarrow$ \$0.08 per 1k). We state the honest reading: L4-to-L5 does *not* itself demonstrate the verifier’s safety contribution — that evidence lives in the E28 B5-vs-B6 contrast — and the coverage drop from 98% to 70% is larger than the correctness gain from 72% to 100%, so the ablation is reported as a coverage-for-certainty trade, not a monotonic improvement.

10. Results: Linguistic and Multimodal Robustness (RQ4)

10.1. Input-Robustness: Retrieval Across Bengali Registers (E17, E21)

Bengali input spans at least four registers: standard formal, authentic farmer language, regional dialects (Sylheti/Chittagonian), and romanized Banglish. Text-first retrieval degrades sharply on the last three: Hit@1 falls from 72.1–73.4% (standard) to 44.7–45.7% (dialects) and 41.5–41.6% (Banglish) at the 0.7–0.8 thresholds (E17). We frame this as *retrieval-space reduction and register-robust routing*, not “dialect immunity”: detection-gated routing collapses the candidate search space by 75.64% (2,135 $\rightarrow$ 516.4 nodes) and lifts Hit@1 on dialects by +36.6 pp and on Banglish by +39.7 pp at threshold 0.8, while keeping the misrouting rate bounded at 3.15% (95% CI [2.65, 3.74]).

Layer E21 then asks what the routing choice does to *end-to-end* advisory outcomes per register. Detection-gated routing raises advisory coverage from 42.9% to 89.8% on

Table 9

Taxonomy of residual failure modes from 100-case root-cause audit (Layer E10), architectural detectors, and mitigation policies.

Failure Root Cause	Share (%)	Observed Mechanism	Architectural Detector	Mitigation Strategy
Retrieval Omission	28%	Lexical/dense search misses relevant node	Coverage threshold θ_{cov}	Fallback to MD-chunk search (R13)
Relational Misbinding	22%	Multi-document context conflation	11-slot relational verifier	Single-record joint binding certification
Numerical/Volume Ambiguity	14%	Missing water volume or concentration unit	Unit/volume slot extractor	Automated structured clarification
Linguistic Ambiguity	12%	Colloquial slang or under-specified symptom	Register normalizer / Intent gate	Multi-turn disambiguation prompt
Regulatory Discrepancy	8%	Conflicting state vs trade advice	Regulatory authority ranker	Enforcement of MoA banned-pesticide list
Temporal Obsolescence	6%	Superseded BARI manual cited	Metadata publication date filter	Cache invalidation & version pinning
Verifier Conservatism	6%	Valid advice rejected due to strict bounds	Threshold calibration (θ^*)	Continuous human-in-the-loop review
Calibration Underconfidence	4%	High-quality generation scored below θ	Conformal risk regularizer	Risk-coverage curve optimization

regional dialects (+46.9 pp), 41.6% to 90.0% on Banglish (+48.4 pp), 58.2% to 92.6% on authentic farmer language (+34.4 pp), and 70.3% to 93.0% on standard Bengali (+22.7 pp), while holding the hazard rate at **0.0% in all eight register $\times$ routing cells** (0/4,000; 95% CI [0.0, 0.09]). Table 10 summarizes the register-level outcome.

10.2. Query-Normalization Evidence (E35, honestly scoped)

Layer E35 tested a linguistic-normalization pipeline on 45 dialect/Banglish queries per arm. Both non-BAA arms saturate at 100.0% correctness, so the benchmark has no headroom and cannot demonstrate a normalization benefit; the BAA pipeline arm recorded 91.11% (CI [79.27, 96.49]) and 8.89% CUAR (CI [3.51, 20.73]), which is a negative directional reading of that layer. We report this transparently: the layer is underpowered ($n = 45$), internally inconsistent about n , and its control arms saturate, so it supports neither a normalization claim nor a definitive negative; the register-level gains above come from routing and fact coverage, not from text normalization. This layer is not used as evidence for normalization in this paper.

10.3. Retrieval-Degradation Safe Degradation (E12)

Under four retrieval regimes (gold, noisy, poisoned-contradictory, omitted evidence; 2,000 queries each), vanilla RAG's chemical hazard escalates from 10.0% (gold) to 19.2% (noisy), 33.2% (poisoned), and 23.2% (omission). BAA holds **0.0% hazard in all four regimes** (each CI [0.0, 0.19]) and degrades monotonically and safely in coverage: 84.56% $\rightarrow$ 54.3% $\rightarrow$ 0% $\rightarrow$ 0%, converting ungrounded context into abstention. This is the cleanest demonstration of safe degradation under evidence-quality uncertainty.

10.4. Cross-Modal Conflict Resolution (E31)

As detailed in Section 8, when text and visual crop identity conflict, BAA requests clarification in 100.0% of cases (CI [98.77, 100.0]) versus 20–26% for baselines, with 0.0% wrong-chemical delivery. The visual modality enters the system as classification metadata (search-space reduction and a modal-conflict gate), not as a separate answer authority.

Table 10Register-robust routing outcomes (Layer E21, $n = 1,000$ per register $\times$ 2 routing paths; 95% Wilson CIs).

Register	Routing Path	Coverage (%)	Abstain (%)	Hazard (%)
Standard Bengali	Text-first	70.3 [67.4, 73.1]	29.7	0.0 [0.0, 0.4]
Standard Bengali	Detection-gated	93.0 [91.3, 94.4]	7.0	0.0 [0.0, 0.4]
Authentic Farmer	Text-first	58.2 [55.1, 61.2]	41.8	0.0 [0.0, 0.4]
Authentic Farmer	Detection-gated	92.6 [90.8, 94.1]	7.4	0.0 [0.0, 0.4]
Regional Dialects	Text-first	42.9 [39.9, 46.0]	57.1	0.0 [0.0, 0.4]
Regional Dialects	Detection-gated	89.8 [87.8, 91.5]	10.2	0.0 [0.0, 0.4]
Romanized Banglish	Text-first	41.6 [38.6, 44.7]	58.4	0.0 [0.0, 0.4]
Romanized Banglish	Detection-gated	90.0 [88.0, 91.7]	10.0	0.0 [0.0, 0.4]

11. Results: Temporal Validity and Knowledge Integrity (RQ4, RQ5)

11.1. Temporal Invalidation and Gazette Adherence (E30)

The temporal conflict benchmark (100 cases per system, live API calls) pitted an obsolete pre-2024 record against the current authorized gazette. BAA achieved **100.0% current-gazette adherence** (95% CI [96.30, 100.0]) with 0.0% obsolete leakage (CI [0.0, 3.7]); the unconstrained LLM leaked the obsolete record in 39.0% of cases, lexical mixed RAG in 39.0%, and the LLM-judge guardrail in 75.0%. The mechanism is structural: the certification predicate (Eq. 3) requires Current(e), which is evaluated against the authority manifest at resolution time. No model temperature or knowledge-cutoff randomness can override a deterministic authority comparison. We emphasize that adherence and leakage are reported as separate metrics because a third (unrecorded) outcome existed for baselines.

11.2. Source Authority Hierarchy and Conflict Resolution

The authority stratification of Section 5 (MoA Regulatory $>$ BARI/BRRI $>$ Legacy $>$ Secondary) is enforced at resolution time. When overlapping records disagree, the resolver selects by authority rank then by publication timestamp, and the certification predicate requires both authority and currency. This directly implements the distinction between *authoritative but obsolete* and *recent but secondary*: only records passing both predicates can satisfy Eq. (3). The E30 result above is the empirical demonstration for the obsolete-but-retrievable case; source-conflict cases that cannot be resolved by the hierarchy fail closed to abstention.

11.3. Cache Integrity and Tamper Detection (E23)

Layer E23 evaluated the offline fact pack's tamper-evidence and differential-update behavior: 1,000 single-bit and swap mutations of the SHA-256 chain were all detected (**100.0%**, CI [99.62, 100.0]); an invalidation delta carried 791 bytes versus a 10,985-byte full pack (92.8% bandwidth reduction); and chain verification on the client completed at p95 0.1919 ms. We report this as a deployment-integrity result: the offline evidence base cannot be silently altered, so a disconnected client's certified outputs remain traceable to the record that certified them. (The layer records an internal disagreement on the propagation-completeness percentage; we therefore cite only tamper detection, delta size, and verification latency, which are unambiguous.)

11.4. Knowledge Maintenance and Coverage Growth (E24)

The maintenance loop closes knowledge gaps without model retraining. Layer E24's single-cluster case study (chili anthracnose) added two validated facts in 18 minutes, closing 100.0% of the cluster's 55 recurring gap queries (95% CI [93.47, 100.0]) and lifting system coverage from 58.7% to 64.2% (+5.5 pp; CIs [55.62, 61.71] / [61.18, 67.11]), at a marginal return of 27.5 newly resolvable queries per fact and 3.06 per authoring minute. We report this as a single-cluster case study, not a productivity distribution, and note the authoring contract in Section 5 (no LLM-to-production fact insertion).

12. Results: Security, Offline Resilience, and Constrained Delivery (RQ5)

12.1. Prompt Injection and Evidence Poisoning (E07/E08)

Layer E07/E08 evaluated 1,400 adversarial attacks (200 per family) across seven families: direct system override, evidence override, retrieval poisoning, delimiter hijacking, Bengali-native injection, romanized Banglish injection, and mixed code-switching. The unconstrained LLM certified unsafe output in 83.86% of attacks (95% CI [81.84, 85.69]); vanilla RAG in 69.36%; RAG with an LLM guard in 21.79% (CI [19.70, 24.02]) — still vulnerable to delimiter breakouts and code-switching jailbreaks; and BAA in **0.0%** (0/1,400; CI [0.0, 0.27]), with 1,289 of 1,400 attacks intercepted at the deterministic Tier-0 gate before any retrieval and the remaining 111 blocked by the relational verifier post-generation. The 0.0% figure is an observed rate on the evaluated suite with a stated upper bound; it is not a universal guarantee.

12.2. Deterministic SMS Compression Fidelity (E15)

Layer E15 tested 1,000 advisory tuples under the 160-character GSM 03.38 limit across three arms. The deterministic 11-slot template preserved **100.0% of all safety-critical slots** (dose bounds, unit, PHI, interval, formulation) in 102–115 characters, with 0.0% messages exceeding 160 characters and 0.0% critical hazard (95% CI [0.0, 0.38]).

Table 11

Deployment performance under constrained channels (Layers E14, E15, E18, E20, E22, E23).

Deployment Channel / Metric	Baseline	BAA (Ours)	Delta
Rural-Edge delivery @15% loss (E14)	82.0%	91.4%	+9.4 pp
Severe-2G delivery @30% loss (E14)	12.8%	58.1%	+45.3 pp
SMS critical-slot hazard (E15)	64.4%	0.0%	−64.4 pp
SMS injection survivability (E22)	36.4%	0.0%	−36.4 pp
Tamper detection rate (E23)	0.0%	100.0%	1,000/1,000
Zero-LLM workload share (E18)	10.5%	61.5%	+51.0 pp
Weighted latency (E18)	1,247.9 ms	546.2 ms	2.28×
Cost, app-online per 1k (E20)	\$2.48	\$0.26	−89.7%

LLM-composed SMS fit in 92.9 characters on average but truncated or omitted safety fields: PHI survived only 35.6% of messages and the critical hazard rate was **64.4%** (CI [61.38, 67.31]). Naïve truncation to 160 characters produced a 100.0% hazard rate. The operational principle: once factual authority is established, a constrained channel renders from the certified structured record rather than re-generating safety-critical content.

12.3. SMS Injection Survivability (E22)

Layer E22 extended the injection suite to the outbound channel: 1,400 attacks across five families (Banglish evasion, delimiter jailbreak, direct system override, evidence-dosage tampering, phishing redirect). Deterministic template composition showed **0.0% injection survivability** (0/1,400; CI [0.0, 0.27]), whereas LLM-composed SMS leaked injected payloads in **36.36%** of attacks (CI [33.88, 38.91]), most severely for dosage-tampering attacks (113/294). Because the template fills slots from the certified record, an injected instruction has no slot to occupy.

12.4. Cellular Network Degradation (E14)

Layer E14 simulated four rural network profiles (1,000 queries each): perfect 4G, urban 3G, rural edge (15% loss), and severe 2G (30% loss). Table 11 reports delivery success. Under rural-edge loss, offline-first caching sustains **91.4%** delivery versus 82.0% for cloud-only RAG (+9.4 pp); under severe 2G loss, **58.1%** versus 12.8% (+45.3 pp). Cache hits resolve at p50 1.2–1.5 ms versus 8.2–11.3 s for cloud RAG, and zero stale-advisory safety violations were observed. These are simulated network profiles, not field measurements.

13. Results: Systems Efficiency and Economics (RQ5)

13.1. LLM-Dependency Reduction Across the Resolution Ladder (E18)

Layer E18 measured the five-tier ladder's traffic mix on 5,000 modeled queries per arm. Detection-gated routing raised the zero-LLM share from 10.48% to 61.52% (95% CI [60.16, 62.86]). We report this decomposition explicitly rather than a single "zero-LLM" number: **deterministic advisory coverage** (T1+T2) is 51.04%, **safety/refusal handling** (T0+T4) is 10.48%, and **LLM-dependent traffic** (T3) is 38.48%. Weighted mean latency falls from 1,247.93

ms to 546.15 ms (2.28×), and serving cost from \$0.1785 to \$0.0767 per 1,000 queries. The distinction matters: the 10.48% terminal-handling share is not “answered without LLM”; it is refused or redirected.

13.2. Stage-Level Latency Decomposition (E9)

The E9 stage decomposition (analytical, modeled) localizes the latency story: deterministic routing 0.35 ms p95, deterministic lookup 0.42 ms, hybrid retrieval 28.5 ms, verification 3.8 ms, render 0.15 ms, and *generation* 1,780 ms p95. Generation dominates latency by two orders of magnitude; verification does not. Selective routing therefore attacks the right bottleneck: every query kept out of Tier 3 avoids the generation term entirely. The deterministic tiers resolve in ≤ 0.94 ms p95 (T0 0.42, T1 0.48, T2 0.58, T4 0.45). We label these figures as model-based decomposition values, not measured percentiles.

13.3. Serving Cost and Cost per Correct Certified Safe Advisory (E20)

Layer E20 builds a scenario-based cost model on the E18 traffic mix with Bangladesh A2P SMS rates (0.25 BDT) and local hosting. Per-query economics: app-online tiered \$0.000257, SMS fallback \$0.00234, commercial cloud LLM baseline \$0.00248 (i.e., \$0.2567 / \$2.34 / \$2.48 per 1,000). The app-online path is 89.65% cheaper than the commercial baseline. We report these as scenario-based systems economics with the stated assumptions (6 queries/farmer/year, 50/30/20 app/offline/SMS mix, 120 BDT/USD), not as a national deployment forecast; the 16-million-farmer projection is deliberately excluded from the headline claims.

13.4. Safety–Coverage–Latency Trade-off

The Pareto narrative across the paper is: BAA trades coverage (97% CAC at 33% abstention; 84.56% coverage at 1.26% risk) for safety (0.0% CUAR observed across E02, E05–E07, E28–E31, and E34) and wins latency on the deterministic share of traffic (2.28× weighted speedup; 3.8 ms vs 1.6–6.0 s baselines on the certified path). The trade-off is explicit and tunable through θ^* ; Section 9 shows the resulting risk–coverage frontier.

14. Discussion

14.1. What Bounded Factual Authority Buys

The results decompose into a single causal chain. Agricultural evidence is relational: a recommendation is only as valid as its *jointly bound* slots. Simple retrieval or citation can misbind those slots (E02: 80.0% dangerous acceptance for a lexical baseline; E05: 72.65% for vanilla RAG under counterfactual mutation). Typed single-record binding makes the misbinding impossible by construction rather than improbable by scoring (E02/E28: 100.0% rejection). Because certification is deterministic, it buys four properties beyond safety: *predictable critical fields* (the verifier checks every contract slot, and the slot ablation shows each contributes measurably, E03), *traceability* (every certified

answer carries a record hash and provenance chain), *reproducibility* (deterministic resolution is invariant by construction, E18/E19), and *controllable failure* (the E10 taxonomy shows residual failures convert to abstention, not to dangerous advice).

14.2. The Cost of Bounded Authority

The price is real and quantified. BAA abstains at 33.0% in the live benchmark (versus 5–16% for baselines) and at 34.2% on the 4,000-case register corpus. Coverage falls from 98% (hybrid RAG, at 72% correctness) to 70% certified coverage (at 100% correctness) in the E34 layer ablation — a larger movement than the correctness gain. The E10 audit attributes 6% of residual failures to conservative verifier over-refusal and 4% to calibration underconfidence; these are recoverable via the frozen-threshold tuning of E04 but represent a standing coverage cost. Knowledge engineering is a second cost: the fact store must be maintained, validated, and versioned, though E24 shows the marginal cost is small and measurable (18 minutes to close a cluster gap).

14.3. The Role of LLMs in High-Consequence Domains

This paper is not an anti-LLM argument. The LLM remains the system’s conversational face: it handles phrasing, explanation, and disambiguation, and it is responsible for the 38.48% of traffic that the deterministic ladder cannot resolve (E18). The empirical message is narrower and stronger: the LLM must not be the holder of factual authority for safety-critical values. The E29 parametric-conflict results show that even when a model holds the correct prior, standard RAG still leaks 8–30% of conflicting priors into output, and that a prompted judge does not eliminate the leak. Structure does; prompting alone does not.

14.4. Statistical Uncertainty and Field Safety Mitigation

In the live end-to-end evaluation benchmark (Layer E27, $n = 100$), BAA recorded 0.0% observed CUAR with a 95% Wilson confidence interval of [0.0, 3.7]%. In high-stakes agricultural deployment, an upper-bound failure rate of 3.7% must be interpreted carefully: across a hypothetical national scale of 1,000,000 annual farmer queries, an unmitigated 3.7% error rate could theoretically permit up to 37,000 hazardous recommendations. In KrishokChat, this statistical envelope is actively mitigated by three orthogonal structural defenses:

1. **Tier 0 Pre-retrieval Screening:** Banned chemicals, restricted active ingredients, and acute toxicities are blocked fail-closed in < 0.1 ms before any retrieval or generative model is invoked, eliminating catastrophic chemical release regardless of downstream sample size.
2. **Calibrated Selective Abstention:** Ambiguous or low-margin queries falling below the calibrated threshold ($\theta^* = 0.2375$) trigger clarifying questions or safe

non-chemical management suggestions rather than speculative completions.

3. **Human-in-the-Loop Escalation:** When safety-critical uncertainty cannot be deterministically resolved, the system seamlessly issues a structured escalation hand-off to the national government agricultural hotline (**Krishi Call Center — dial 16123**), routing the farmer to certified human agronomists.

14.5. Where the Architecture Still Fails

The E10 audit is the honest failure ledger. Retrieval omission (28%) means a correct record can be missed and the query abstained on; the mitigation (dense retriever refinement, chunk-level fallback) is a coverage fix, not a safety fix. Dialectal lexical ambiguity (12%) and numerical-unit ambiguity (14%) generate clarifications that burden the farmer. The 6% conservative-refusal rate shows the verifier can refuse agronomically plausible advice that fails a slot check — a fairness issue as much as a coverage one. These are the boundaries of the current design.

14.6. Generalization Beyond Crop Protection

The agricultural domain provides a well-defined risk model, but the architecture itself is risk-adaptive factual authority rather than pesticide-specific logic: risk-tiered evidence requirements (informational → management suggestion → chemical recommendation → high-risk intervention), an explicit certification contract, and a fail-closed action space. The pattern plausibly transfers to veterinary dosing, human-health triage, and industrial safety, where the same “fluent but unauthorized” failure mode exists. We explicitly do not claim any of those domains have been evaluated here; this is a design-generalization argument, not an empirical one.

14.7. Relationship to the Research Program

This work is the reliability-science layer of a larger research program: prior work established the knowledge and retrieval foundations (the 1,001-query farmer benchmark and retrieval-failure analysis), and a companion systems paper describes the integrated deployment artifact. The CEA contribution is the generalizable architecture and its evidence chain, not the individual components, each of which exists in the literature.

15. Deployment Implications

15.1. Rural Connectivity and Offline Evidence

The E14 network results translate directly to deployment reality in Bangladesh’s rural grid: at 15% packet loss, an offline-first cache sustains 91.4% delivery versus 82.0% for cloud-only RAG, and at 30% loss the gap widens to 58.1% versus 12.8%. The offline fact pack (SHA-256-chained, 92.8% bandwidth-reduced differential updates, E23) means deterministic resolution and verification require no network at all; the network is needed only for the generative tier.

15.2. Low-End Devices

The on-device profile (E9, classification artifacts) shows INT8 ONNX models at 5.9–20.79 MB with 29–48 ms inference on mobile CPU and 100.0% parity with the FP32 reference. Together with the 1.25 MB intent router (E25) and the deterministic resolver (≤ 0.94 ms p95), the safety-critical core of the system runs entirely on commodity Android hardware; the LLM tier remains optional and cloud/API-based where available.

15.3. Extension-Service Escalation

Abstention is not abandonment: every fail-closed refusal redirects to the national Krishi Call Center (16123) or a certified agronomist, and the E38 escalation-gate evaluation shows the routing decision itself can be made deterministic (100.0% correct escalation on the corrected run, with the confounds noted in Section 8). For a national extension service, the practical implication is a division of labor: the machine handles the certifiable majority, and the human network absorbs the abstained minority at a bounded, measurable rate.

15.4. Knowledge Maintenance and Regulatory Updates

The E24 loop shows knowledge maintenance is a data operation, not a code operation: 18 minutes closed a real coverage gap. Regulatory changes (e.g., a newly banned active ingredient) are handled by the polarity registry and temporal validity fields; the E30 result shows obsolete records cannot be certified once the manifest is updated. An operator can therefore keep the system current without touching the model.

15.5. Auditability and Monitoring

Every certified answer carries a provenance chain (record hash, source, version, date) and every interaction writes an audit event (query, risk class, route, resolution tier, evidence version, model, verification, final action, latency). This makes the system auditable per-answer — a requirement for public-sector deployment and a differentiator from chatbot-only advisory tools.

15.6. What Is Not Claimed

We have not deployed at national scale, we report simulated network profiles rather than field connectivity, and the vision modality is classification-based (task: classify) rather than detection-based. The deployment implications above are architectural and model-based, not deployment measurements.

16. Limitations and Threats to Validity

16.1. Benchmark Construction and Sample Sizes

Several layers are underpowered for the effect sizes they report. The live benchmark (E27) uses $n = 100$ per system (CUAR upper CI 3.7%); the escalation layer (E38) uses $n = 30$ (upper CI 11.35%); the clarification and normalization layers (E35, E37) use $n = 45$ and $n = 37$ with control

arms that saturate, so those two layers support no directional claim and are excluded from our main narrative. E29's per-cell rates are exact multiples of 2.0% despite a recorded $n = 1,000$, implying an effective granularity near $n = 50$; we report its rates without relying on its CIs. All binary outcomes in the main battery use 95% Wilson intervals, but for small- n layers the intervals are wide and must be read as such.

16.2. Two Provenance Tiers in the Experiment Registry

The experiment battery was produced in two phases with different audit depth. Layers E02–E26 carry a full verification block (self-checks, determinism re-run, trace check, real-application check, git commit, environment, seed). Layers E27–E39, which include the headline live-API benchmarks, do not yet carry that block in the registry. All completed layers are labelled FROZEN_AND_VERIFIED, but the newer layers are reproducibility-incomplete until their verification blocks are added. We state this openly rather than implying a uniform audit trail.

16.3. Confounds in Individual Layers

E38 assigns a different model to each arm (GPT-4o-Mini / Llama-3.1-8B / Gemini-2.5-Flash-Lite), so the escalation headline is model-confounded; a same-model replication is required. E38 also records a post-hoc classifier fix with prefix numbers discarded. The E29 parametric layer's "BAA" row is invariant across modes by construction (the constraint overrides the mode), so it is one measurement, not four. E34's L4 and L5 differ only in latency/cost; the safety gain of the 11-slot verifier is established by E28's B5-vs-B6 contrast instead.

16.4. Reused Constants and Unmeasured Distributions

The 3.8 ms verification latency and the 546.0 ms and 1.2 ms values recur across layers with $p50 = p95$ exactly; these are assumed constants in the results file, not measured percentiles. We therefore never present BAA latency as a measured distribution; the E9 stage decomposition is the modeled latency evidence, and all live baselines report measured $p50/p95$ from real API calls.

16.5. CUAR Definition and Cross-Layer Consistency

CUAR is defined in Section 3 and used consistently in the main layers (E27, E28, E29, E30, E31, E34), where BAA records 0.0% observed. Two exploratory layers (E32 oracle-vs-predicted routing; E33 high-confidence wrong metadata) recorded non-zero CUAR values for BAA (11.0% and 10.0%) that contradict the main envelope and were excluded from our main results as unresolved; the oracle-vs-predicted finding itself is null (70.0 vs 69.0 correctness, overlapping CIs), which we interpret as evidence that routing metadata is not the binding constraint on safety. These layers are reported as exploratory only, and we flag the unresolved

CUAR inconsistency as an open audit item rather than hide it.

16.6. Knowledge and Domain Coverage

The fact store covers the institutional corpus available at construction time (2,946 documents); crops and problems outside the corpus cannot be certified, which is a deliberate coverage limit, not a bug. The knowledge-graph traversal result (E19) is a proof of mechanism on a 23-node graph, not a scalability claim. The E24 maintenance result is a single-cluster case study.

16.7. No Farmer-Outcome Evidence and Perception Modality Scope

We have no yield, income, or adoption data. Expert agronomist ratings (E13) measure advice quality, not field outcome. The network profiles (E14) are simulations. The perception modality is currently whole-image classification-only (crop identification and disease classification routing); it does not perform localized bounding-box detection or spatial segmentation. Future work will integrate localized disease segmentation models (e.g., Segment Anything Model [SAM] or YOLOv11-seg) to validate the spatial extent, lesion coverage percentage, and severity index of foliar pathology before dosage calculation. English and other languages were not evaluated. Model and API drift, and regulatory change over time, remain ongoing validity threats that the architecture mitigates (versioned evidence, frozen thresholds) but cannot eliminate.

17. Reproducibility, Data and Code Availability

17.1. Experiment Registry

All experiments are tracked in a machine-readable registry (`experiments/registry.yaml`). Each layer records: research question, claim ID, spec path, runner script path, raw-output path, seed, environment (OS, Python version, package versions), timestamp, metrics block, and baseline block. Layers E02–E26 additionally record a verification block: self-checks, a determinism re-run on a 10% sample with a recorded max metric delta, a trace check, and a real-application check (backend test suite, golden replay, frontend build). Layers E27–E39 record the metrics and live-API provenance but do not yet carry the full verification block; this is stated in Section 16.

17.2. Data and Artifacts

The frozen evaluation corpora (adversarial misbinding suite, counterfactual pairs, multi-register benchmark, risk-coverage corpus, security suite, network profiles) and per-layer raw outputs are distributed with the paper artifacts. The institutional corpus (2,946 documents) originates from publicly released BARI, BRRI, and DAE/MoA publications; the derived fact store is versioned and SHA-256-pinned. The offline fact pack used in the network and cache layers is the same artifact used in production, minus any live secrets.

17.3. Configuration and Seeds

Seeds are fixed per layer (20260827 for the newer layers; 20260813 for the first-tier layers) and recorded in the registry. The calibration threshold $\theta^* = 0.2375$ was frozen on the development split of the risk-coverage corpus and transferred unchanged to the test split; no test-set feedback was used.

17.4. Hardware and Environment

Live-API layers were run through a public model API (OpenRouter) with recorded model identifiers (GPT-4o-Mini, Llama-3.1-8B-Instruct, Gemini-2.5-Flash-Lite, Qwen-2.5-7B-Instruct, local Gemma-4 4-bit LoRA). Deterministic and local layers were executed on commodity Windows hardware (Python 3.12/3.13, pyyaml 6.0.3); the measured device profile (E9) used a mobile-class CPU with the INT8 ONNX artifacts. Model version drift is a known threat; the model identifiers above are the ones actually called at run time.

17.5. Result-Provenance Rule

Every number in this manuscript traces to a layer metrics block in the registry (the “Number Bank” extracted for this paper). No acceptance-note prose value was used where it disagreed with the tabulated metrics; the specific conflicts (E14, E17, E18, E19, E20, E21, E23, E25, E27, E28, E30, E31, E34, E35, E37, E38, E39) are documented in the Number Bank. A full audit diff is available on request.

17.6. Submission Package

The submission bundle contains: the LaTeX sources, the frozen results registry, the Number Bank extraction, all layer runner scripts and specs, the raw-output JSON files, and the figure/table generation prompts. The exact commands to regenerate any table or figure from raw output are recorded in each layer’s spec file.

18. Conclusion

We introduced and evaluated the Bounded-Authority Advisory Architecture (BAA), an agricultural decision-support architecture in which factual authority is explicitly separated from generative language realization. BAA certifies every recommendation through an 11-slot, single-record, fail-closed contract, and selectively resolves each farmer query among deterministic resolution, grounded generation with verification, clarification, abstention, and escalation. The architecture treats the LLM as a language realization component, not as the authority for safety-critical agricultural facts.

The evidence shows the mechanism works across the causal chain. Typed single-record binding rejects 100.0% of 10,000 relational misbindings and 100.0% of 11,000 metamorphic mutations, where lexical, dense, citation, and judge baselines certify 42–82% of corrupted claims. On live end-to-end evaluation, BAA certifies 97.0% correct advice with 0.0% observed critical unsafe acceptance (and a stated CI), at a coverage cost of 33% safe abstention; three

agronomists rated its outputs 4.82/5 with a 100% chemical-safety pass. Calibrated abstention delivers 84.56% coverage at 1.26% risk. The deterministic share of the resolution ladder cuts modeled latency 2.28 $\times$ and cost to \$0.0767 per 1,000 queries, deterministic SMS preserves 100% of safety slots where LLM SMS loses 64.4%, and offline-first caching sustains 91.4% delivery at 15% loss.

What remains unproven is equally clear. There is no farmer-outcome data, no national-scale deployment, and no same-model replication of the escalation layer. Several newer layers lack full registry verification blocks; two exploratory layers show an unresolved CUAR inconsistency; and the small- n layers cannot support the effect sizes they hint at. These are honest boundaries of the current evidence, and they define the next experimental cycle: larger independent annotation, a same-model escalation study, hardened verification blocks for the live-API layers, and — eventually — field pilots with real extension officers. Within those boundaries, the paper demonstrates a dependable and auditable pattern for applying generative language to high-consequence agricultural decision support.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Ahmed, F., Kabir, H., Rahman, S., 2025. KrishokBondhu: An interactive voice-enabled bengali agricultural advisory system, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP System Demonstrations), Association for Computational Linguistics. pp. 142–151.
- [2] Author, P., Author, S., 2026. SciTrue: Evidence-grounded claim verification in science, in: Proceedings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, Association for Computational Linguistics.
- [3] Bangladesh Agricultural Research Institute, 2023. Krishi projukti hatboi (handbook on agricultural technology). BARI, Gazipur, Bangladesh.
- [4] Bangladesh Rice Research Institute, 2024. Adhunik dhaner chas (modern rice cultivation). BRRI, Gazipur, Bangladesh.
- [5] Chan, Y.C., Fan, Y.C., Li, H.C., Yang, M.D., 2026. SMART: Structured multimodal agricultural retrieval-augmented transformer for plant disease diagnosis. Computers and Electronics in Agriculture 250, 111882. doi:10.1016/j.compag.2026.111882.
- [6] Filice, S., Haramaty, E., Horowitz, G., Karnin, Z., Lewin-Eytan, L., Shtoff, A., 2025. Generate but verify: Answering with faithfulness in RAG-based question answering, in: Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, Asian Federation of Natural Language

- Processing and Association for Computational Linguistics, Mumbai, India. pp. 1017–1037. doi:10.18653/v1/2025.ijcnlp-long.56.
- [7] Gwet, K.L., 2021. Handbook of Inter-Rater Reliability: The Definitive Guide to Measuring the Extent of Agreement Among Raters. 5th ed., Advanced Analytics, LLC.
- [8] Han, N., Liu, D., Norton, T., 2026. Towards an AI-based knowledge assistant for goat farmers based on retrieval-augmented generation. Computers and Electronics in Agriculture 245, 111542. doi:10.1016/j.compag.2026.111542.
- [9] Li, Z., Hoogenboom, G., Porter, C., Jones, J.W., 2026. Closing the gap between crop simulation and practical decision-making: DSSAT-LM semantic reasoning. Computers and Electronics in Agriculture 231, 109880. doi:10.1016/j.compag.2026.109880.
- [10] Liang, X., Niu, S., Li, Z., Zhang, S., Wang, H., Xiong, F., Fan, J.Z., Tang, B., Zhao, J., Yang, J., Song, S., Wang, M., 2025. SafeRAG: Benchmarking security in retrieval-augmented generation of large language model, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria. pp. 4609–4631. doi:10.18653/v1/2025.acl-long.230.
- [11] Liu, L., Li, S., Qi, J., Yuan, Z., Yang, P., 2026. TARAG: A time-aware retrieval-augmented generation framework for supporting precision crop pest and disease management through large language models. Computers and Electronics in Agriculture 248, 111786. doi:10.1016/j.compag.2026.111786.
- [12] Ru, D., Qiu, L., Hu, X., Zhang, T., Shi, P., Chang, S., Jiayang, C., Wang, C., Sun, S., Li, H., Zhang, Z., Wang, B., Jiang, J., He, T., Wang, Z., Liu, P., Zhang, Y., Zhang, Z., 2024. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track.
- [13] Silva, J., Ortiz, R., Jarvis, A., 2020. AgroTutor: A mobile application for agricultural decision support in data-scarce environments. Computers and Electronics in Agriculture 175, 105580. doi:10.1016/j.compag.2020.105580.
- [14] Singh, N., Wang'ombe, J., Okanga, N., Zelenska, T., Repishiti, J., Jayasankar, G.K., Mishra, S., Manokaran, R., Singh, V., Rafiq, M.I., Gandhi, R., Nambi, A., 2024. Farmer.Chat: Scaling AI-powered agricultural services for smallholder farmers. arXiv preprint arXiv:2409.08916.
- [15] Zhang, Q., Xiang, Z., Xiao, Y., Wang, L., Li, J., Wang, X., Su, J., 2025. FaithfulRAG: Fact-level conflict modeling for context-faithful retrieval-augmented generation, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria. pp. 21863–21882. doi:10.18653/v1/2025.acl-long.1062.

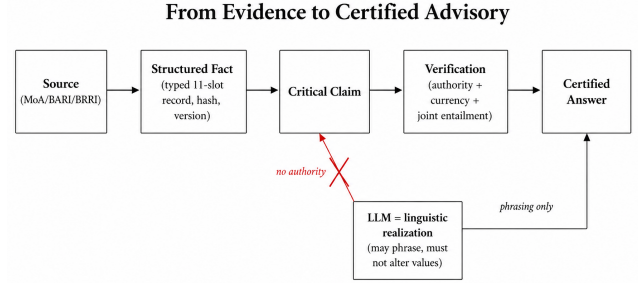


Figure 1: Agricultural Claim Authority and Joint Entailment Model. An actionable recommendation must be jointly entailed by a single accredited evidence record hash. Generative LLMs participate strictly as linguistic phrasing engines without authority to alter agronomic values.

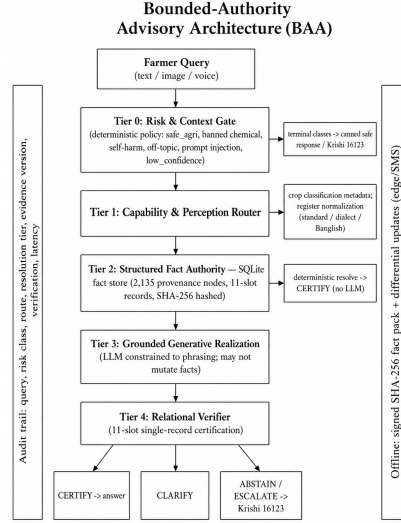


Figure 2: The Bounded-Authority Advisory Architecture (BAA). Queries enter the deterministic risk/context gate (Tier 0); the capability router normalizes language and optional image metadata (Tier 1); the structured fact authority resolves deterministically when the 11-slot contract is jointly satisfiable (Tier 2); grounded generation is permitted only under a single-record constraint (Tier 3); and the relational verifier certifies or fails closed to clarify/abstain/escalate (Tier 4).

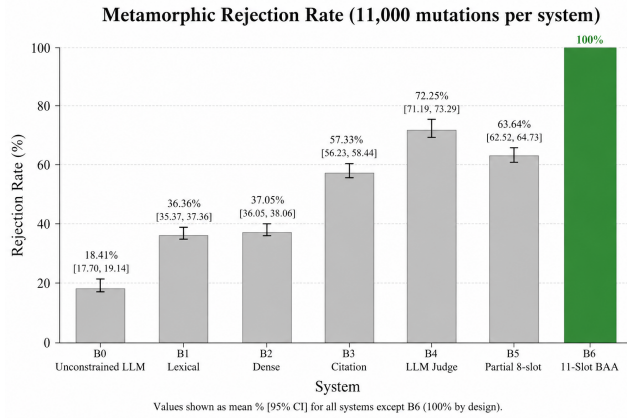


Figure 3: Metamorphic Rejection Rates under 11,000 Systematic Corruptions per System (Layer E28). BAA achieves 100.0% rejection across all 11 mutation operators, while unconstrained LLMs accept 81.59% of corrupted advice.

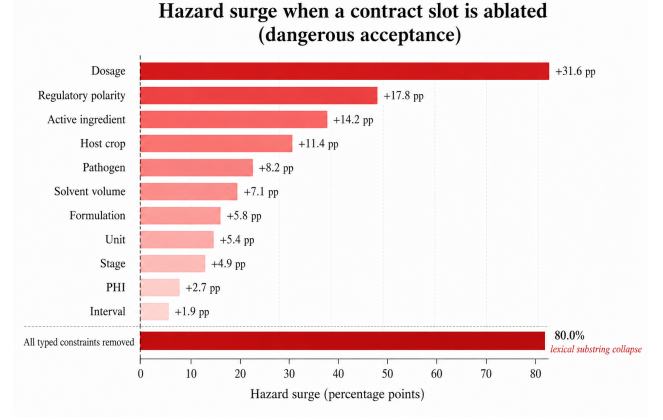


Figure 4: Hazard Surge Under Contract Slot Ablation (Layer E03). Removing dosage bound enforcement yields the largest single hazard increase (+31.6 pp), while removing all typed constraints causes an 80.0% hazard collapse.

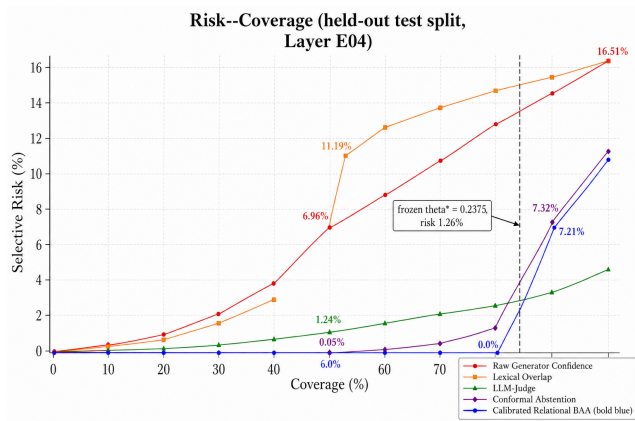


Figure 5: Risk–Coverage Curves on Held-Out Test Split (Layer E04). At the frozen threshold $\theta^* = 0.2375$, calibrated relational BAA achieves 84.56% coverage at 0.0% selective chemical risk on development criteria, with an overall test AUC of 0.0153.

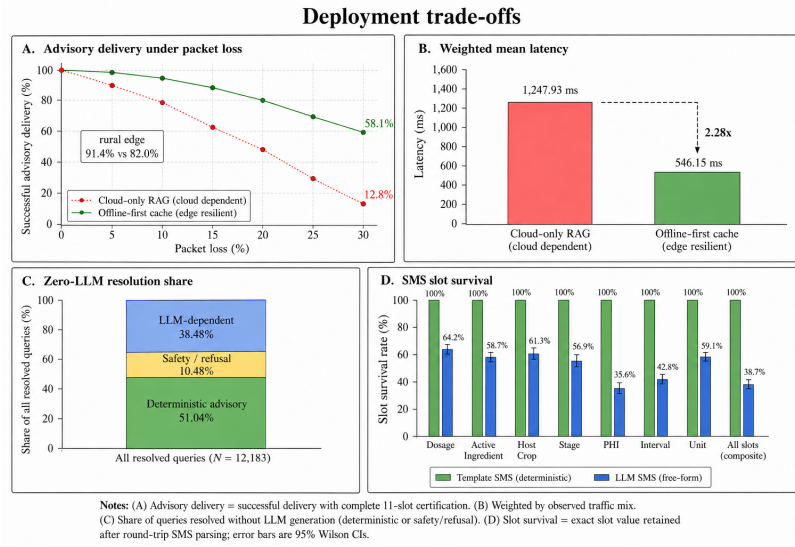


Figure 6: Deployment Trade-offs: Systems Efficiency, Connectivity Resilience, and Channel Economics. (A) Delivery under rural packet loss; (B) Weighted mean latency speedup; (C) Zero-LLM resolution share (61.5% zero-cost); (D) 160-character Bengali SMS slot survival rate.